The Agentic Decade Starts with Open Infrastructure

Mohamed Awad

Executive Vice President and General Manager of Cloud AI Business Unit, Arm



OCP GLOBAL SUMMIT

October 12-15, 2026
San Jose, California

AI INFRASTRUCTURE GROWTH

220 GIGAWATTS

Global data center capacity demand by 2030.
~40% growth since 2025 projection

OCP MOMENTUM

700 members

Worldwide by the end of 2026.
40% growth since 2025

In one year...

$7 TRILLION

Estimated global spending on AI Datacenters and Infrastructure by 2030

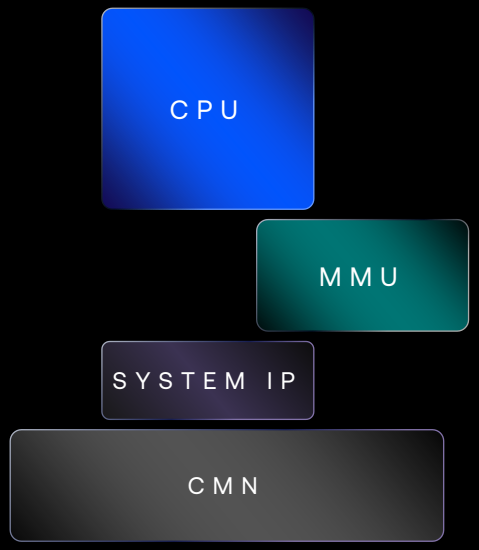
Source: McKinsey & Company research, 2026

ARM EVOLUTION

Arm AGI CPU

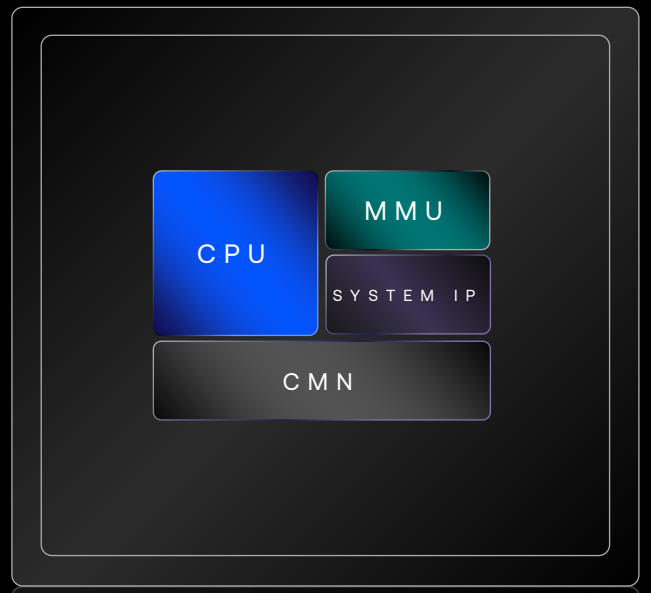
Shipping production silicon alongside IP/CSS/Chiplets.

Arm: Accelerating deployments for the ecosystem



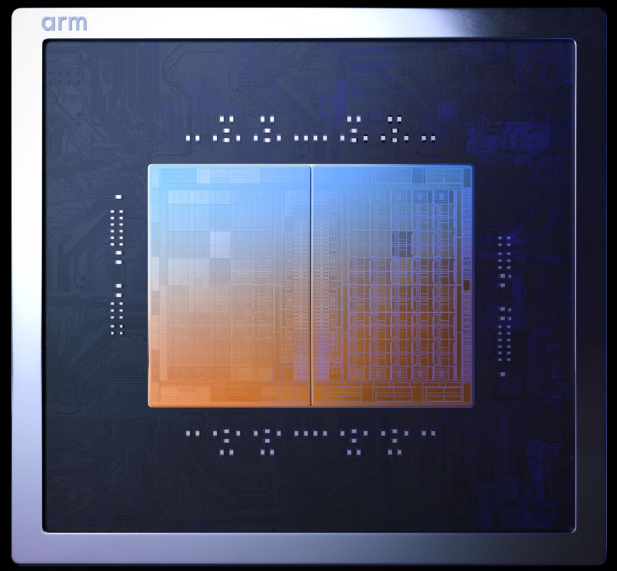
arm neoverse

IP



arm neoverse

Compute Subsystems (CSS)



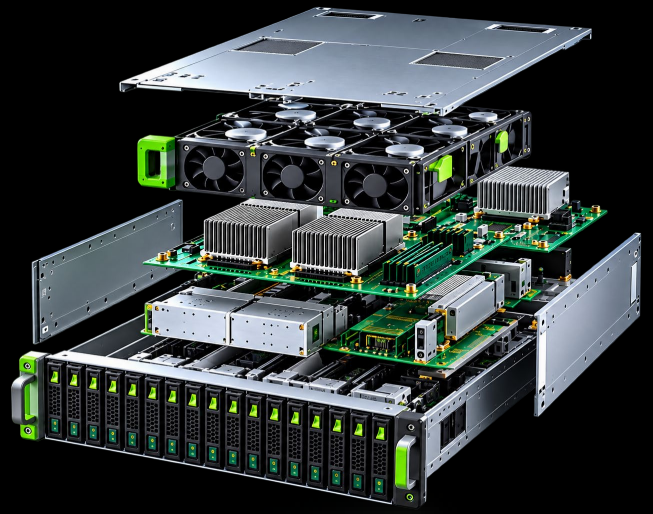
arm AGI CPU

Silicon

OCP: Scaling innovation for the community



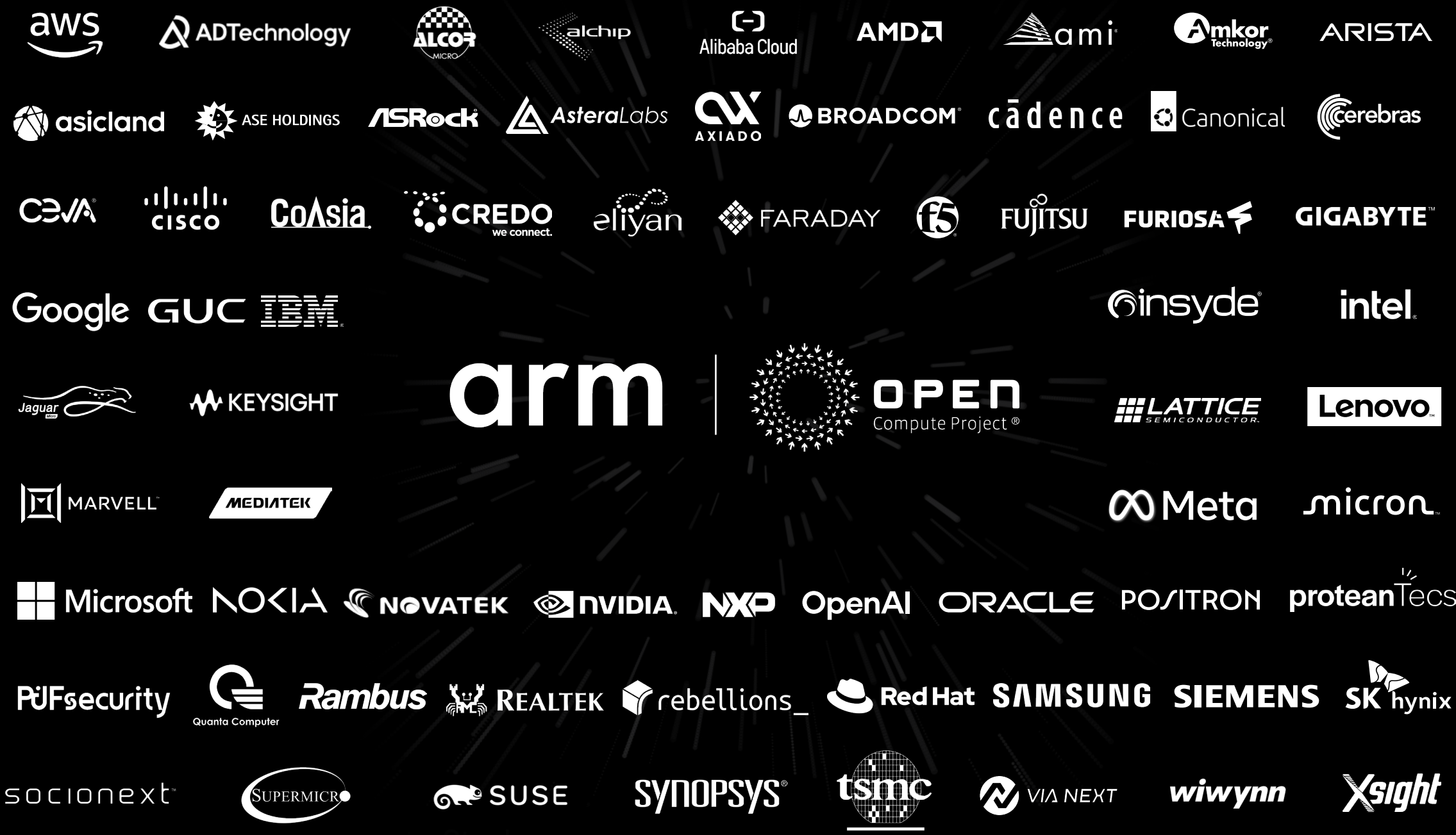
Specifications



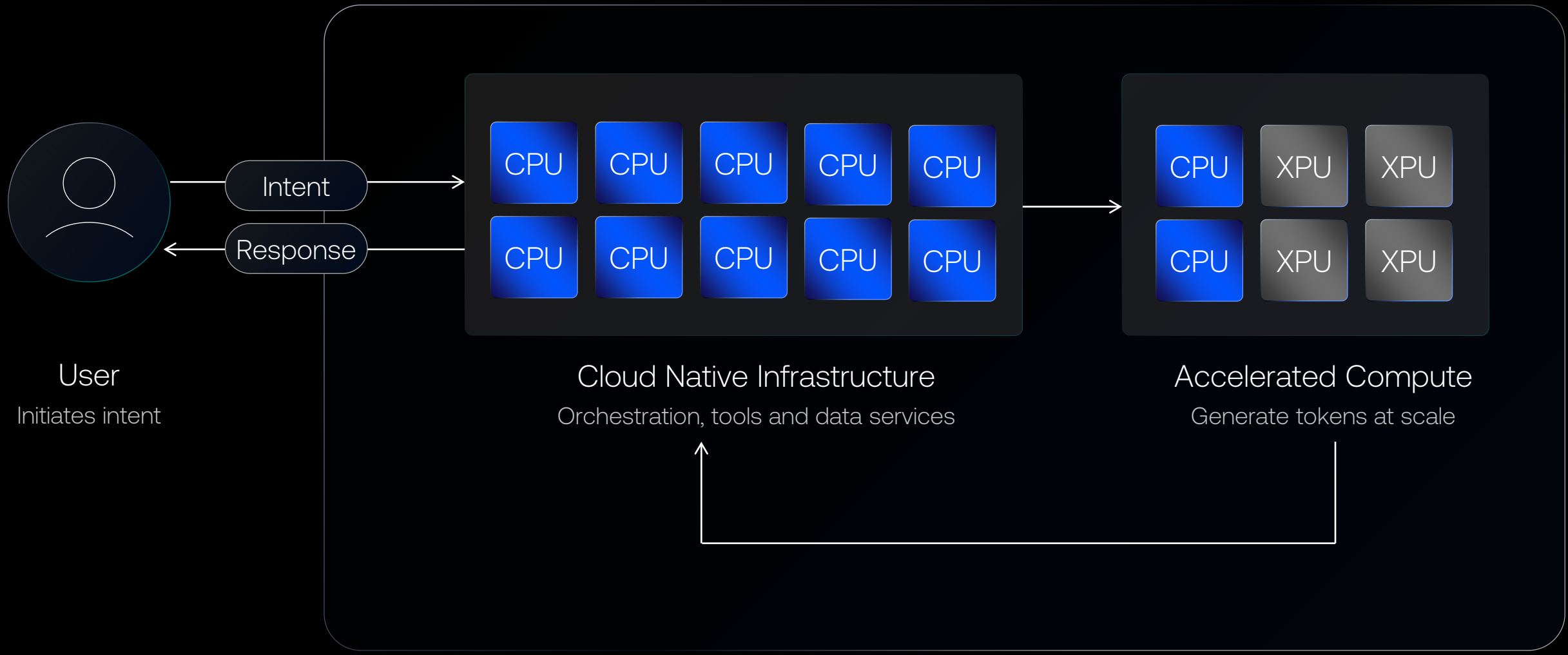
Reference Designs



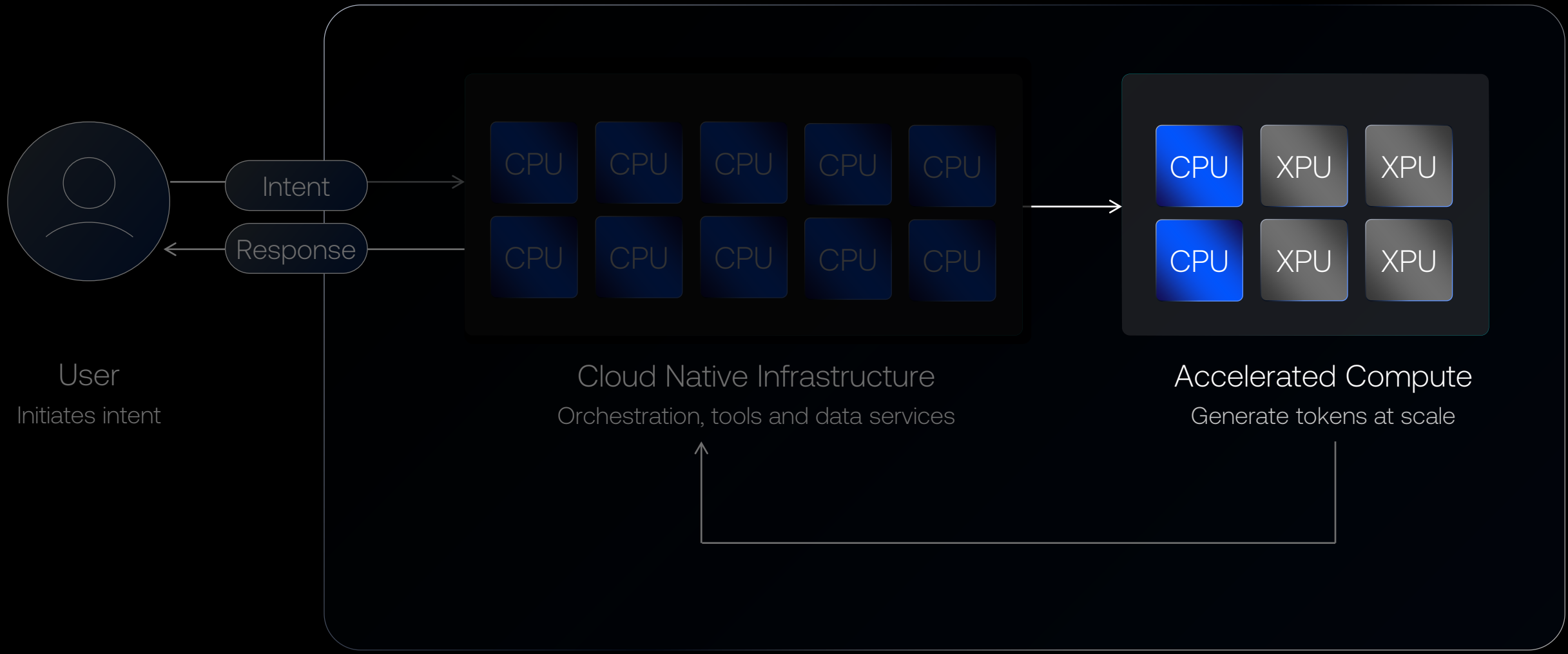
Marketplace Products


aws
ADTechnology
ALCOR MICRO
Alchip
Alibaba Cloud
AMD
ami
Amkor Technology
ARISTA
asicland
ASE HOLDINGS
ASRock
AsteraLabs
AXIADO
BROADCOM
cadence
Canonical
cerebras
CEVA
cisco
CoAsia
CREDO
we connect.
eliyan
FARADAY
f5
FUJITSU
FURIOSA
GIGABYTE
Google
GUC
IBM
insyde
intel
Jaguar
KEYSIGHT
arm
OPEN
Compute Project ®
LATTICE
SEMICONDUCTOR
Lenovo
MARVELL
MEDIATEK
Meta
micron
Microsoft
NOKIA
NOVATEK
NVIDIA
NXP
OpenAI
ORACLE
POSITRON
proteanTecs
PUFsecurity
Quanta Computer
Rambus
REALTEK
rebellions_
Red Hat
SAMSUNG
SIEMENS
SK hynix
socionext
SUPERMICRO
SUSE
SYNOPSYS
tsmc
VIA NEXT
wiwynn
Xsight

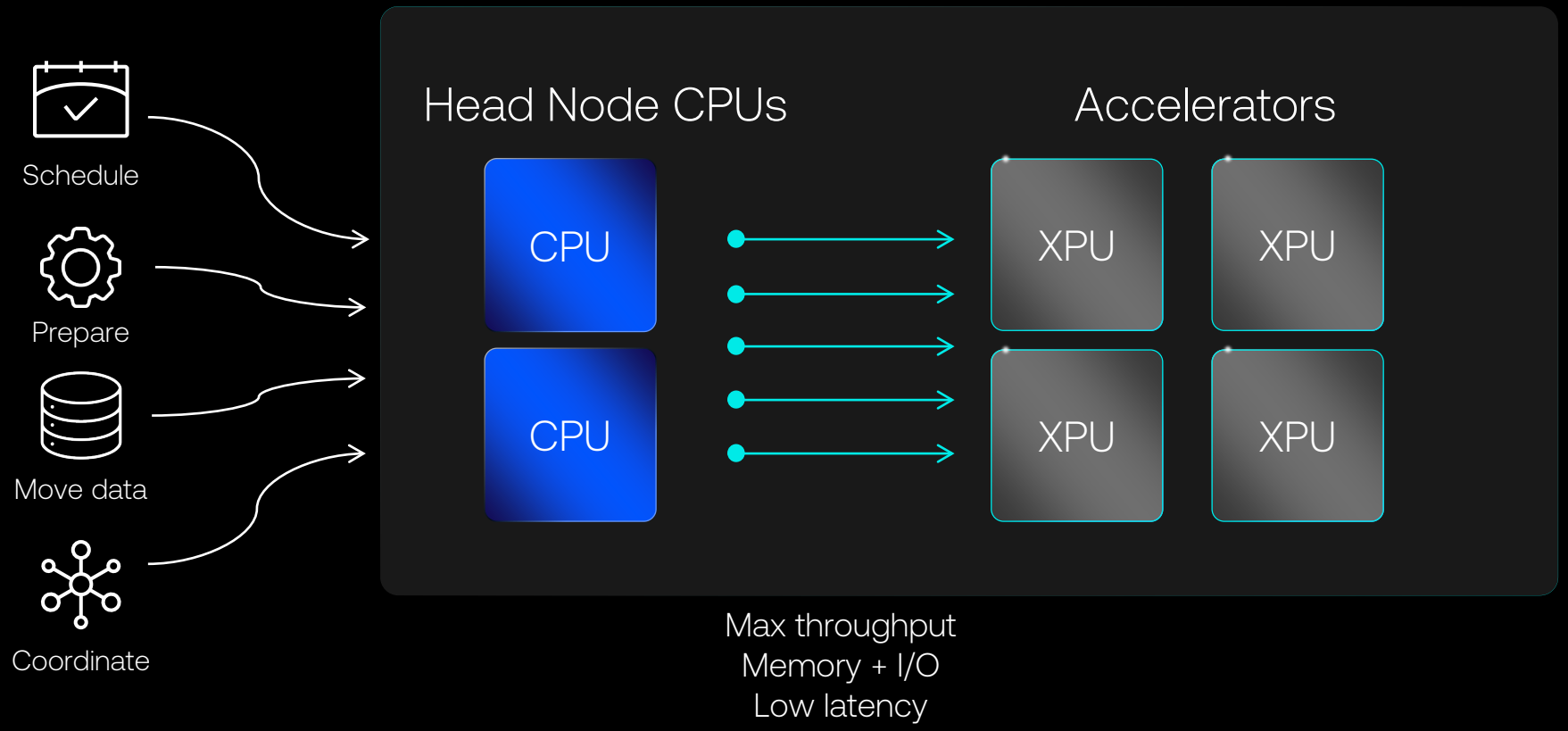
The AI datacenter



The AI datacenter



The accelerator should never wait on the CPU



aws



GRAVITON + TRAINIUM

Google Cloud



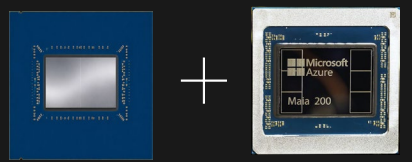
AXION + TPU

Meta



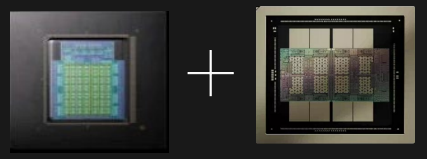
AGI CPU + MTIA

Microsoft Azure



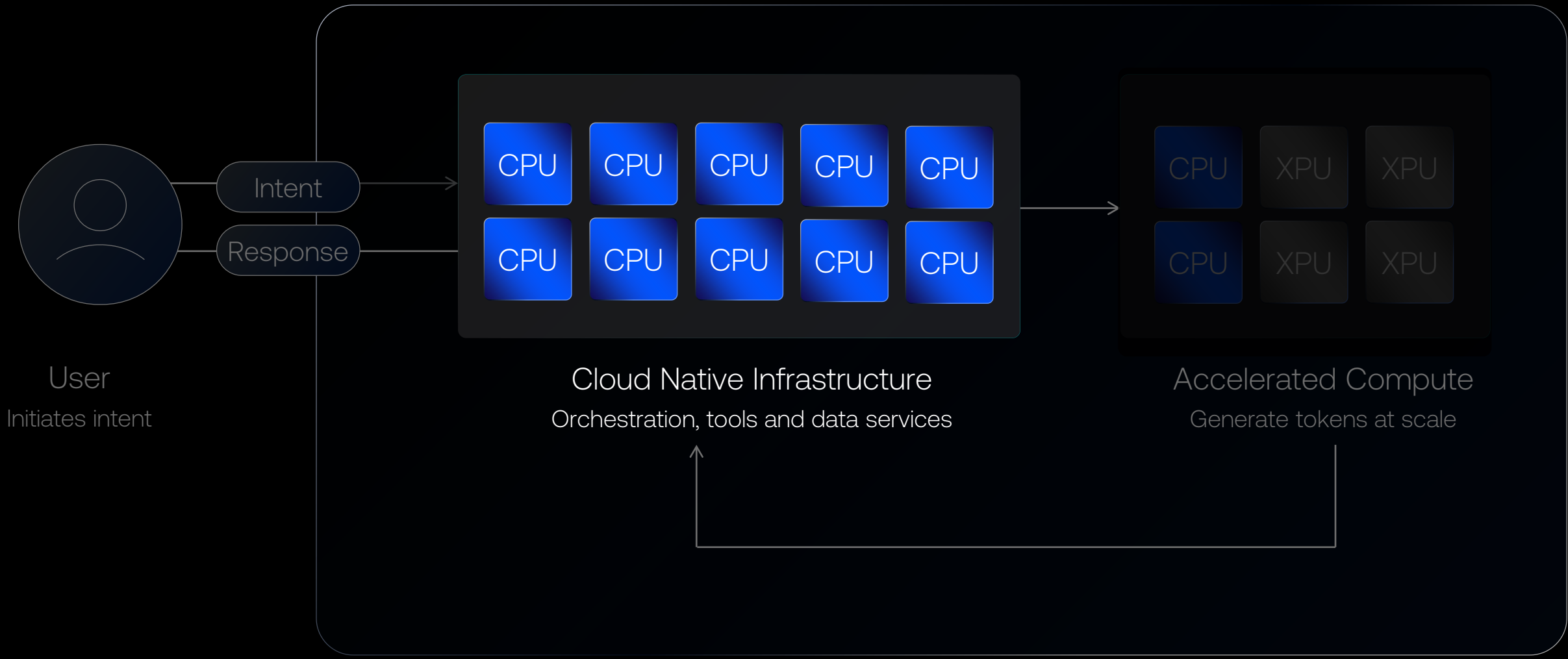
COBALT + MAIA

NVIDIA

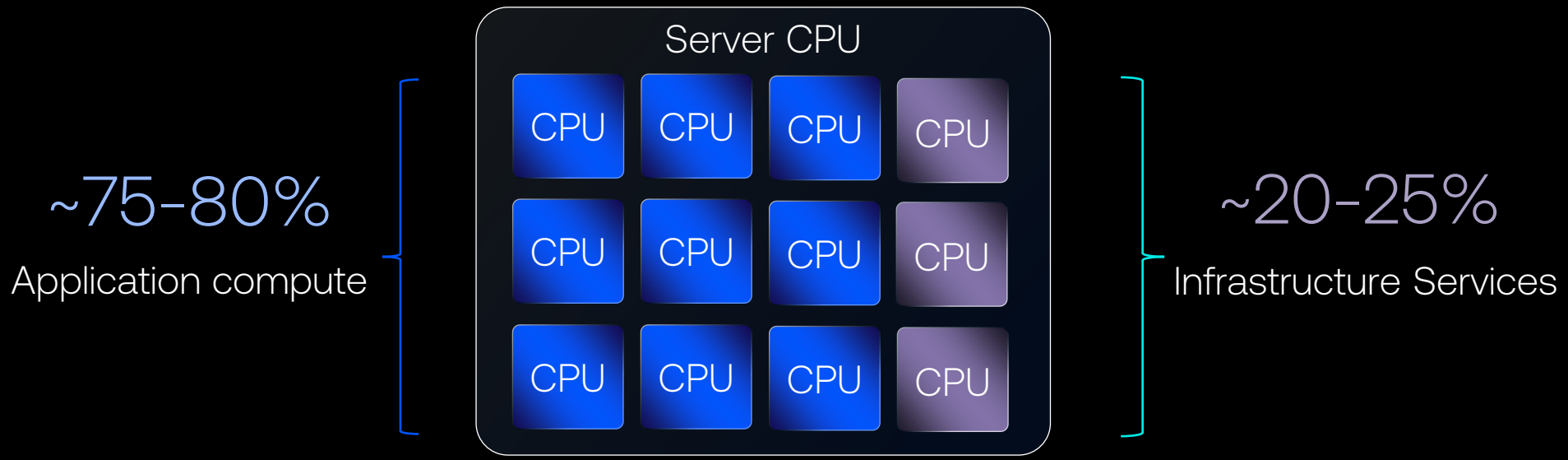


VERA + RUBIN

The AI datacenter

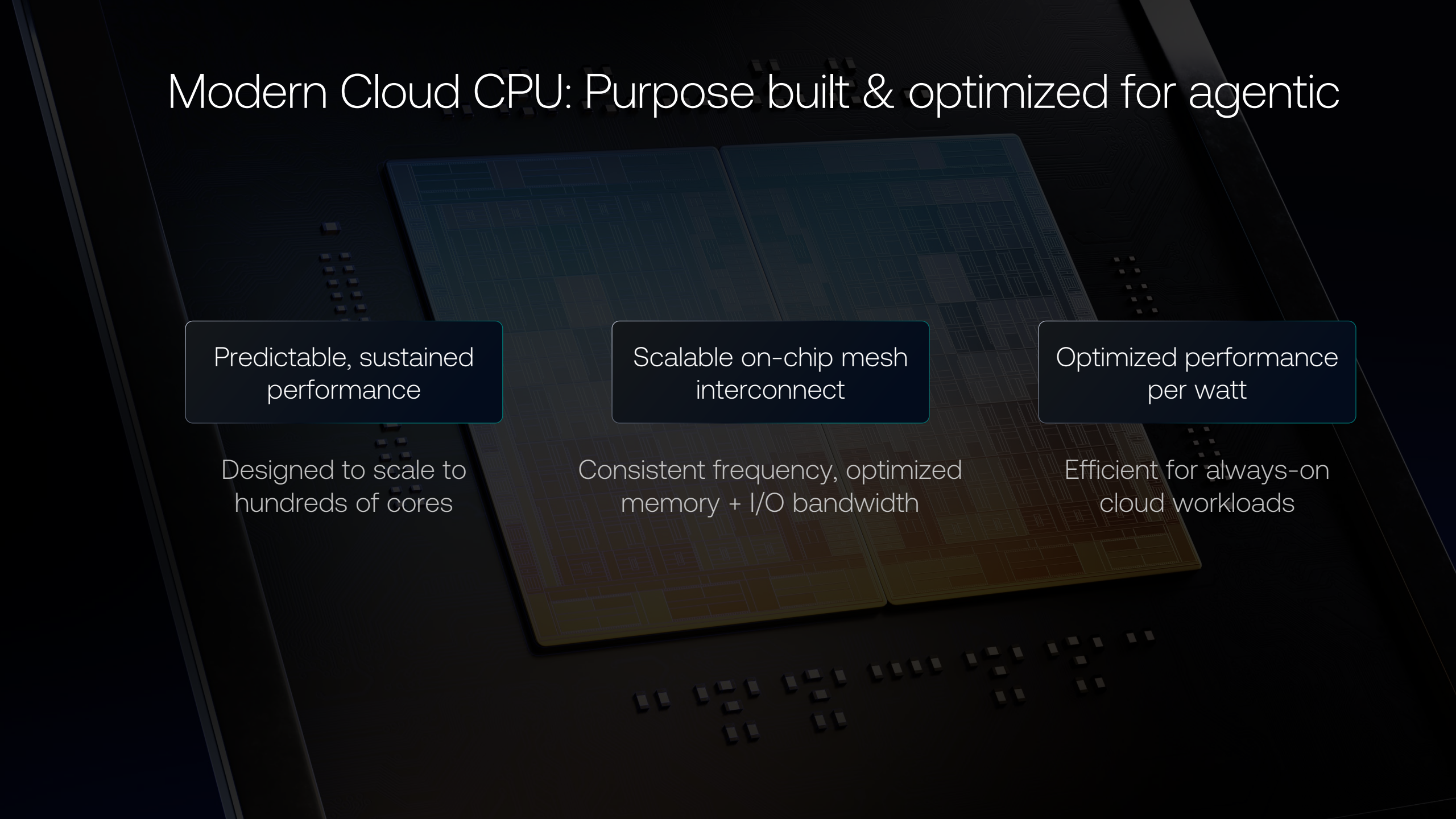


Legacy cloud - Apps and infra services ran on legacy CPUs





Modern Cloud CPU: Purpose built & optimized for agentic



Predictable, sustained performance

Designed to scale to hundreds of cores

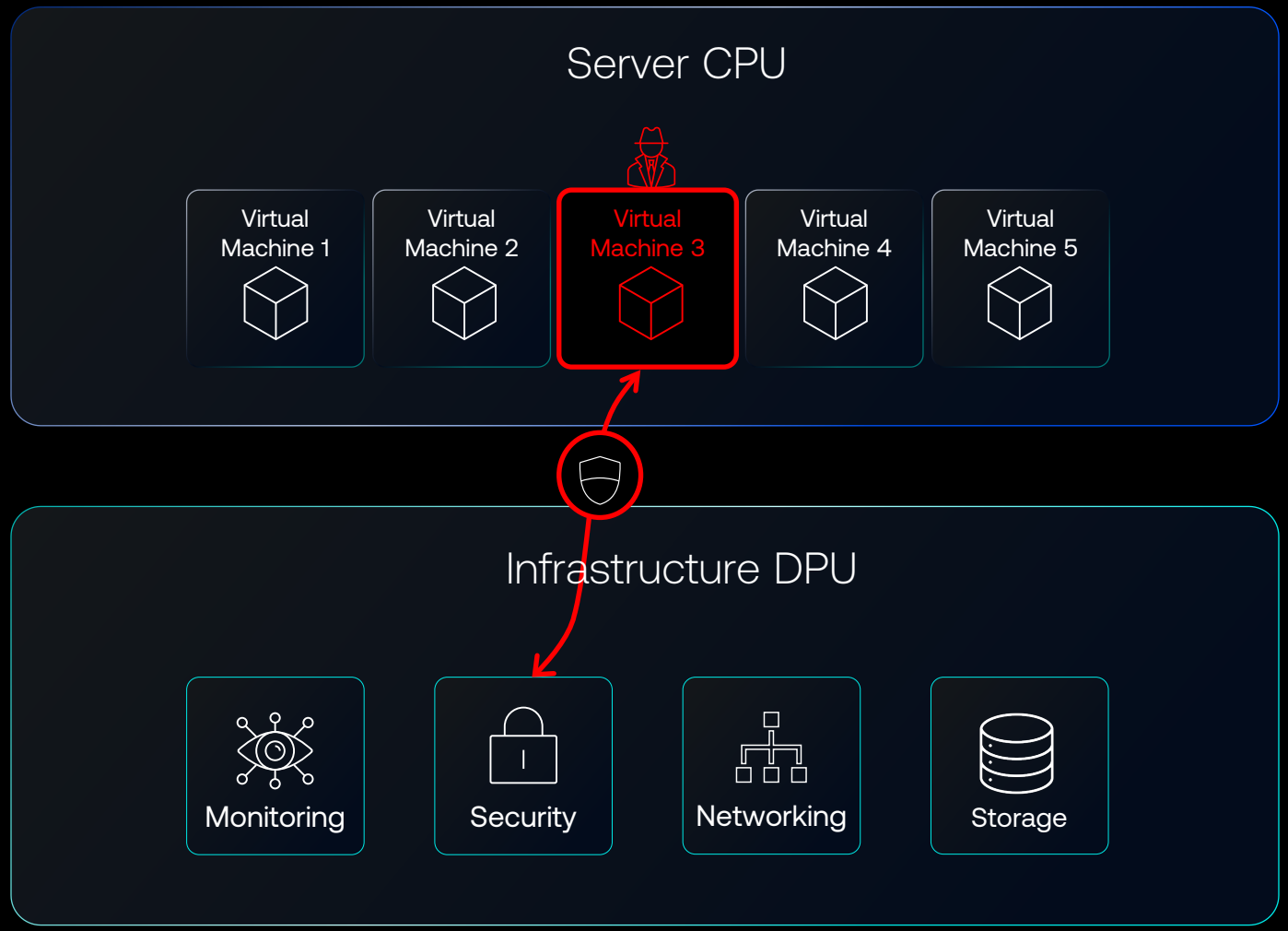
Scalable on-chip mesh interconnect

Consistent frequency, optimized memory + I/O bandwidth

Optimized performance per watt

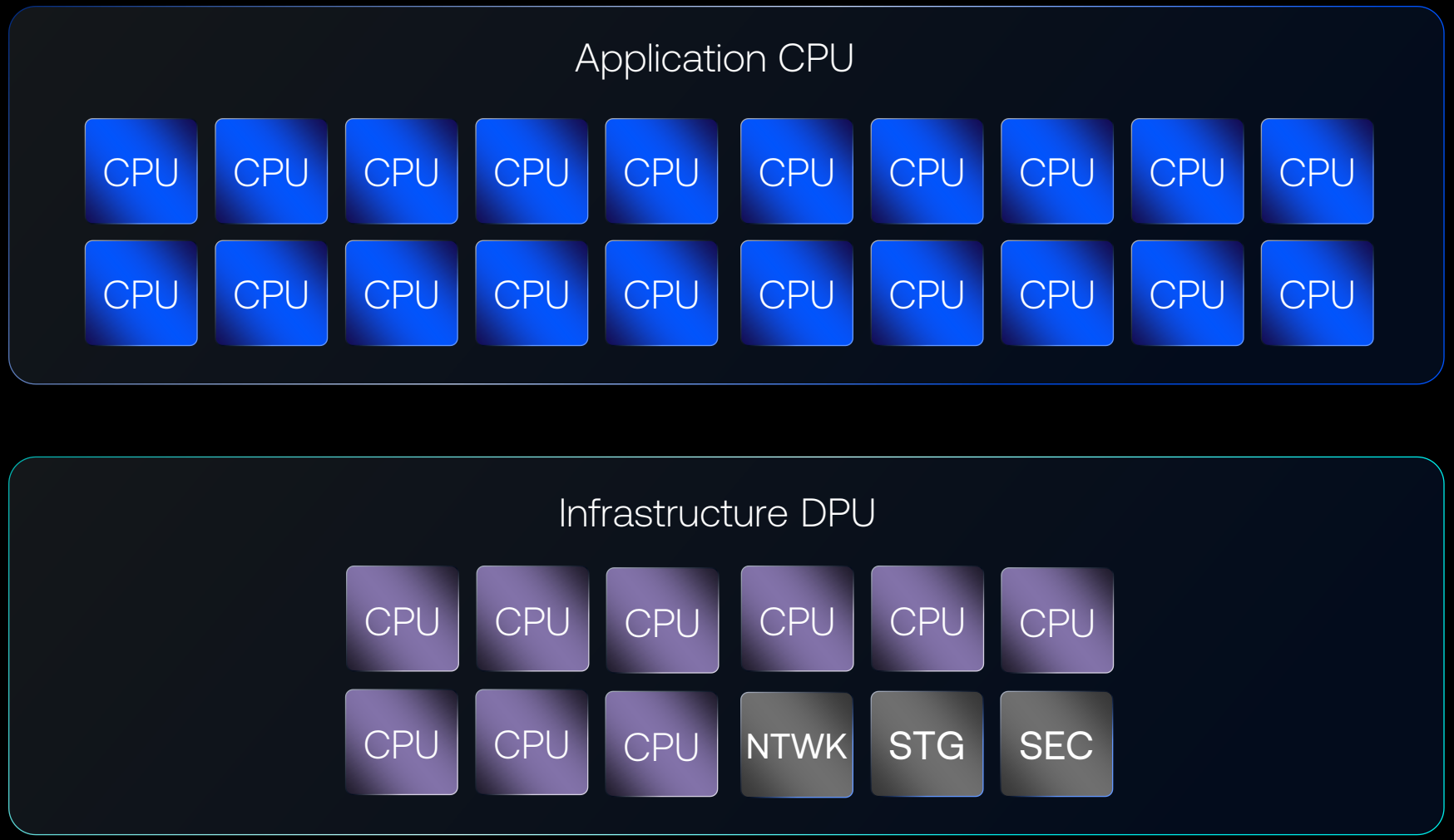
Efficient for always-on cloud workloads

Strong isolation and security enforcement with DPUs



Cloud Providers separated compute from infra services with DPUs

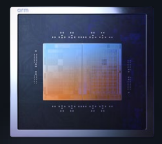
Increased core count, Improved security and performance



- Deterministic Performance
- Secure VM Isolation
- Optimized for Perf/watt

Arm Based CPUs

AGI CPU



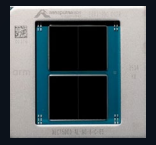
arm

Yitian 710



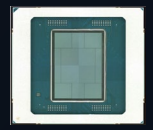
Alibaba Cloud

Graviton



aws

Monaka



FUJITSU

Axion



Google Cloud

Cobalt



Microsoft Azure

Vera



NVIDIA

Arm Based DPUs

Nitro



aws

Salina



AMD

E2100



intel

JM800



云豹智能

OCTEON 10



MARVELL

Azure Boost



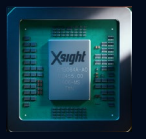
Microsoft Azure

Bluefield4



NVIDIA

E1



Xsight

Agentic AI is becoming THE infrastructure workload.

tom's HARDWARE

7.3Trillion agent tokens vs. 1.4Trillion human tokens on OpenRouter in August 2026

Oct 2, 2026

Forbes

Meta's Muse AI Assistant Hits 5 Million Downloads - Outpacing Growth Of ChatGPT, Grok And Claude

Sep 30, 2026

Bloomberg

SpaceXAI's Grok Bot Agent Tops 400,000 Users After First Month

Sep 22, 2026



Personal agents



Coding agents

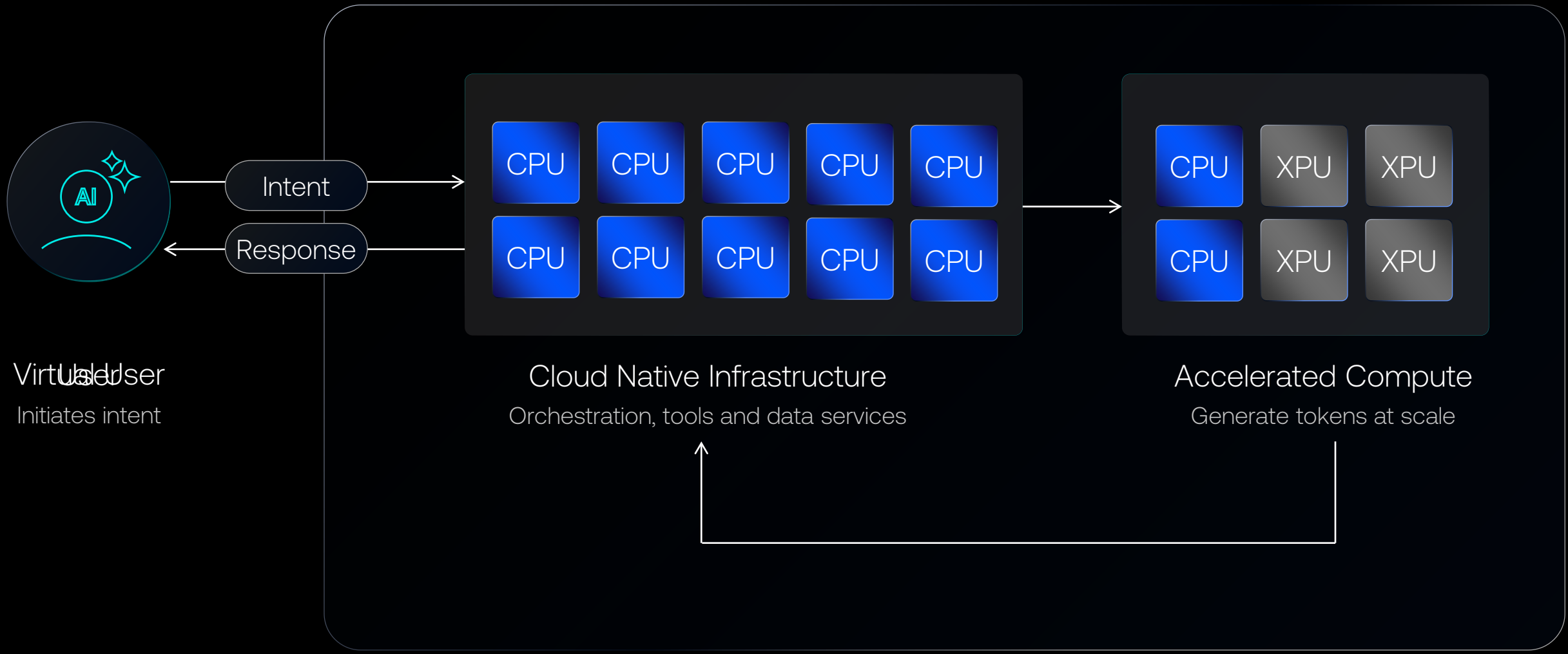


Enterprise agents

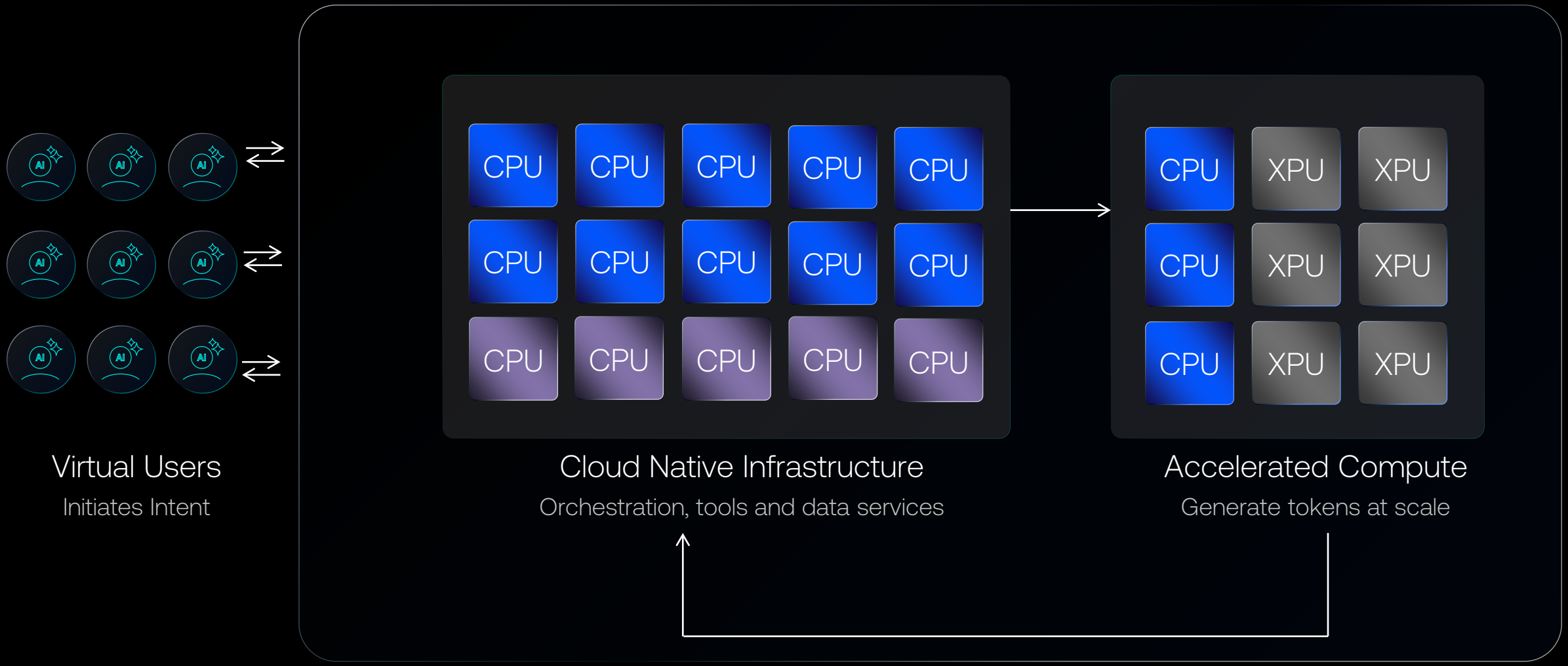


Research agents

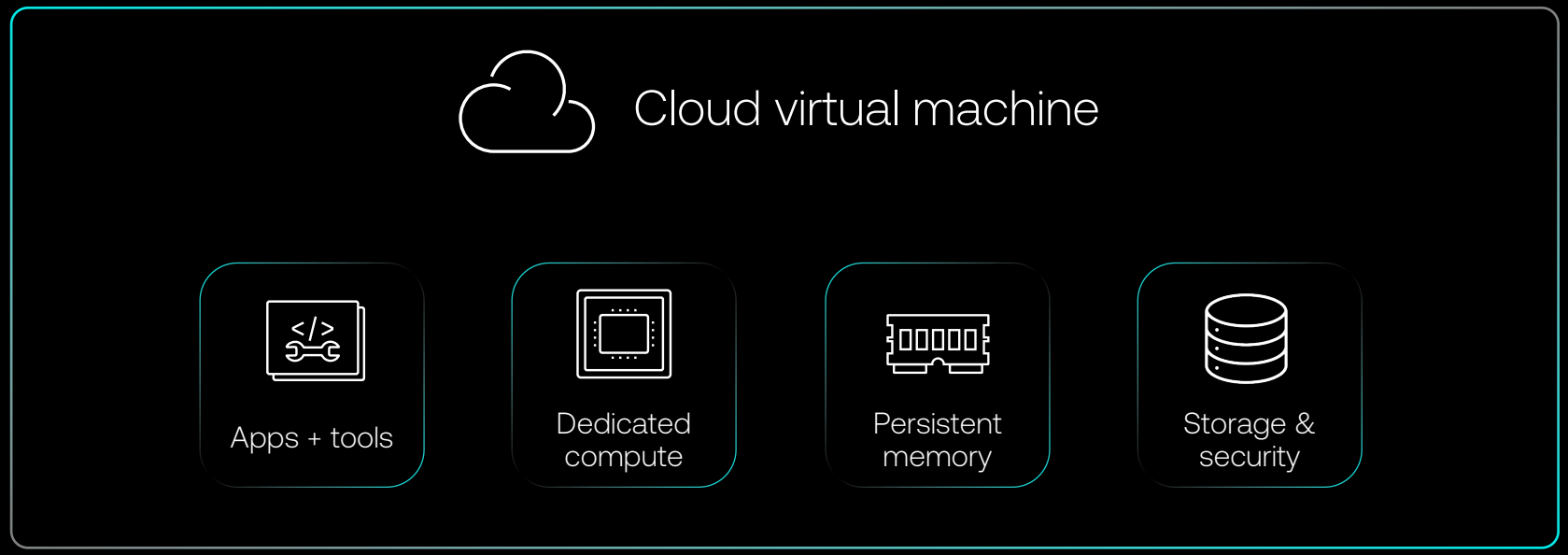
The AI datacenter



The AI datacenter



And every user has a virtual machine in the cloud

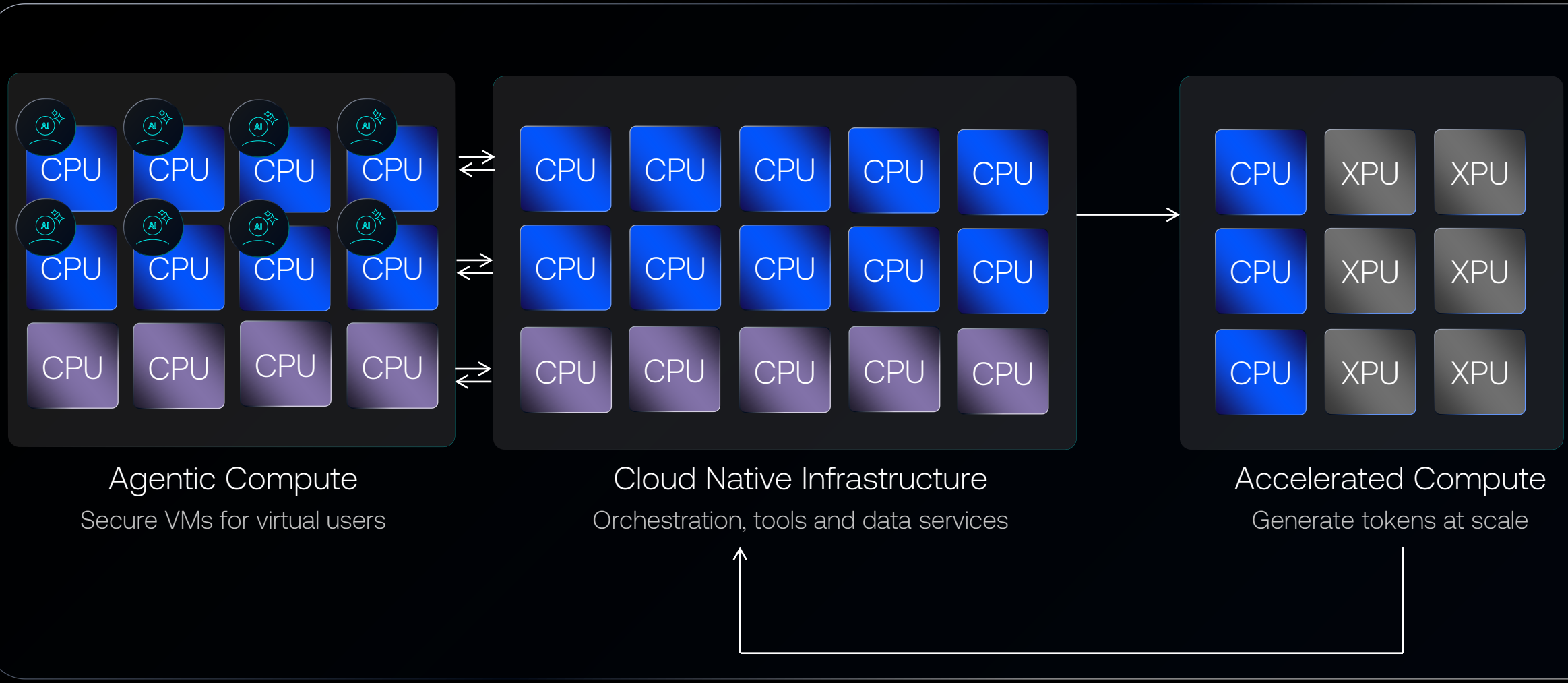


Meta Muse
Secure cloud virtual machine

OpenAI Dots
Per agent cloud container

GrokBot
Persistent cloud computer

The AI datacenter for Agents



One AI data center. Three very different CPU jobs.

AI Head Node

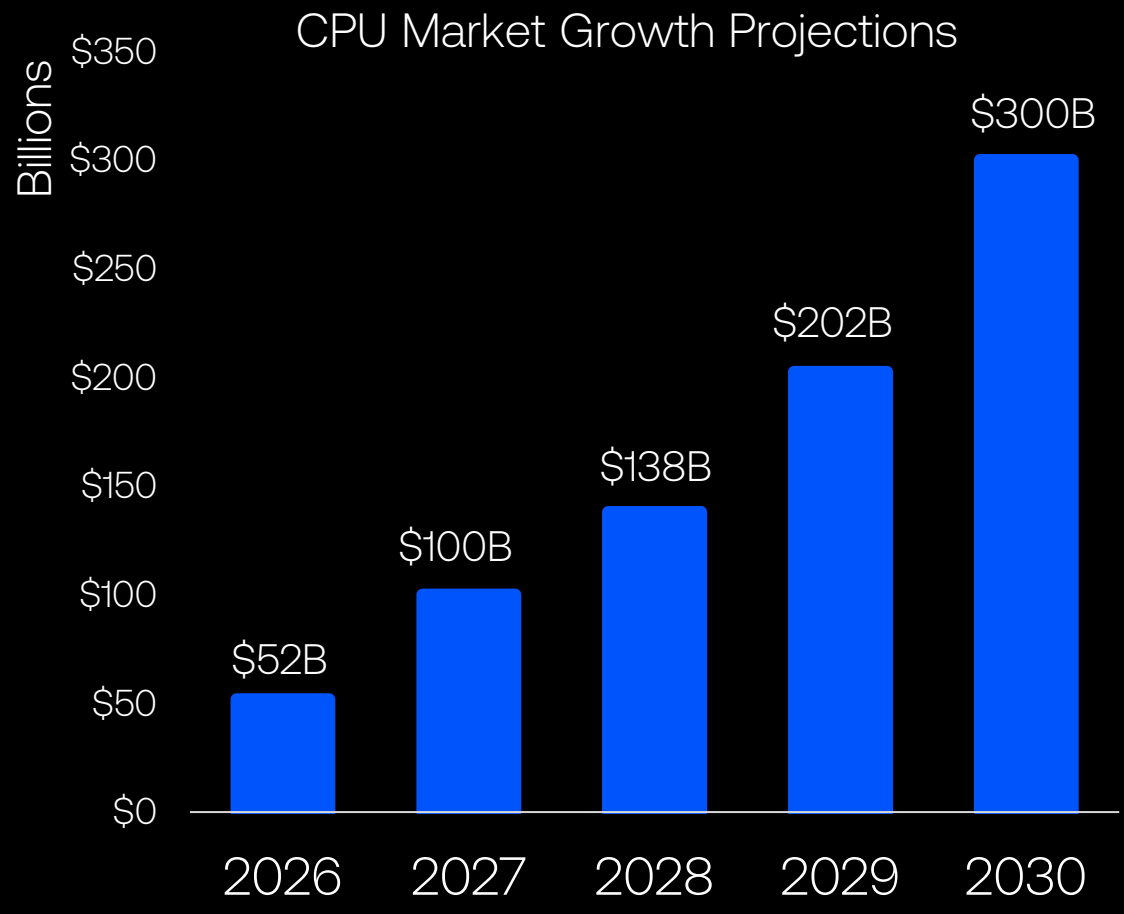
Feed the accelerator

Cloud Native

Serve every user

Agentic

Virtual Users



Source: Citi Research, Mercury Research 2026

One AI data center. Three very different CPU jobs.

AI Head Node
Feed the accelerator

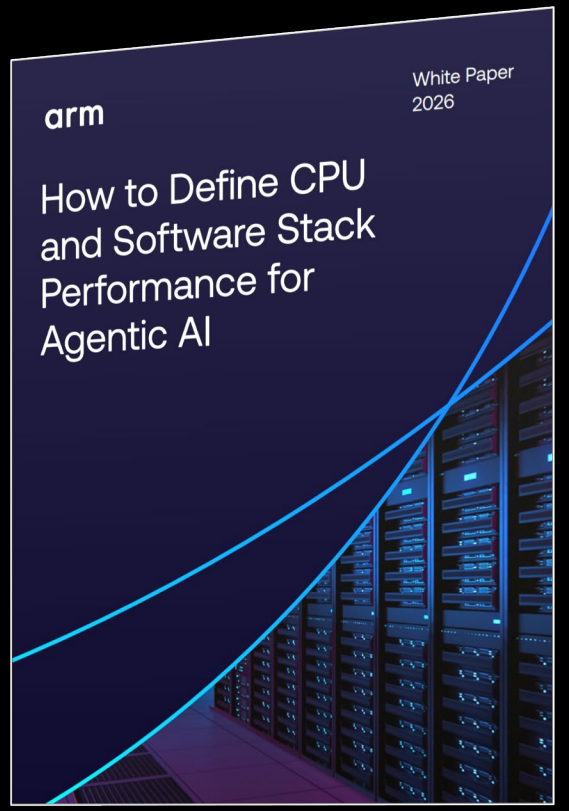
Cloud Native
Serve every user

Agentic
Virtual Users

In collaboration with

FUJITSU Meta OpenAI

Access Whitepaper



Scaling open infrastructure innovation with OCP

From open specifications to rack-ready infrastructure

SERVER STANDARDS

SBSA · SBBR · SBMR

Standards for Arm ecosystem servers

COMPLIANCE/ SECURITY

ACS · ADAC

Architecture compliance for Arm servers

RELIABILITY/ SERVICEABILITY

RAS API

In collaboration with Microsoft

CHIPLET FOUNDATION

FCSA Spec

Open chiplet design used in AGI CPU

REFERENCE SERVER

Arm AGI CPU

Reference design for OCP

RACK FOUNDATION

ORv3 & ORW Racks

In collaboration with Meta for AGI CPU



Build the next generation of AI infrastructure. Together.

Shape the open infrastructure for the agentic era.

- XPU/ ACCELERATOR
- MEMORY/ KV CACHE
- SYSTEMS/ RACKS
- FABRIC, NETWORKING
- STORAGE & I/O
- FIRMWARE MANAGEMENT
- COOLING – AIR/LIQUID
- POWER/ MANAGEMENT
- FACILITIES/ DATACENTER




arm
OPEN
Compute Project ®